```
> restart: with(Groebner) :

  d := 17;

  degre := d :

  SS := 1 * x^floor(d/2);

  RR := 1 + z^(d) + x^(d) + y^(d) + SS :

  R := expand(T^(d) * subs({x = X/T, y = Y/T, z = Z/T}, RR));

  RT := diff(R, T);
  RX := diff(R, X);
  RY := diff(R, Y);
  RZ := diff(R, Z);

  Ideal := {R, RT, RX, RY, RZ};

  Variables := {T, X, Y, Z} :
```

$$d := 17$$

$$SS := x^8$$

$$\boldsymbol{R} := T^{17} + T^9\ X^8 + X^{17} + Y^{17} + Z^{17}$$

$$RT := 17\ T^{16} + 9\ T^8\ X^8$$

$$RX := 8\ T^9\ X^7 + 17\ X^{16}$$

$$RY := 17\ Y^{16}$$

$$RZ := 17\ Z^{16}$$

$$Ideal := \{17\ Y^{16},\ 17\ Z^{16},\ 8\ T^9\ X^7 + 17\ X^{16},\ 17\ T^{16} + 9\ T^8\ X^8,\ T^{17} + T^9\ X^8 + X^{17} + Y^{17} + Z^{17}\} \quad \textbf{(1)}$$

```
> ##Smoothness

  HilbertDimension(Ideal, {T, X, Y, Z});

  Solve(Ideal, tdeg(Variables));
```

$$\mathbf{0}$$

$$\{[[T,\ X,\ Y,\ Z],\ plex(Z,\ Y,\ X,\ T),\ :\ ]\} \quad \textbf{(2)}$$

```
> ##Intersection of four divisers

  HilbertDimension({R, RT, RX, RY}, {T, X, Y, Z});

  HilbertDimension({R, RT, RX, RZ}, {T, X, Y, Z});

  HilbertDimension({R, RT, RY, RZ}, {T, X, Y, Z});

  HilbertDimension({R, RX, RY, RZ}, {T, X, Y, Z});
```

```
HilbertDimension({RT, RX, RY, RZ}, {T, X, Y, Z});
```

$$0$$
$$0$$
$$0$$
$$0$$
$$0 \qquad (3)$$

> ***##Intersection of three divisers***

```
HilbertDimension({R, RT, RX}, {T, X, Y, Z});
HilbertDimension({R, RT, RY}, {T, X, Y, Z});
HilbertDimension({R, RT, RZ}, {T, X, Y, Z});
HilbertDimension({R, RX, RY}, {T, X, Y, Z});
HilbertDimension({R, RX, RZ}, {T, X, Y, Z});
HilbertDimension({R, RY, RZ}, {T, X, Y, Z});
HilbertDimension({RT, RX, RY}, {T, X, Y, Z});
HilbertDimension({RT, RX, RZ}, {T, X, Y, Z});
HilbertDimension({RT, RY, RZ}, {T, X, Y, Z});
HilbertDimension({RX, RY, RZ}, {T, X, Y, Z});
```

$$1$$
$$1$$
$$1$$
$$1$$
$$1$$
$$1$$
$$1$$
$$1$$
$$1$$
$$1 \qquad (4)$$

> ***##Intersection of tow divisers***

```
HilbertDimension({R, RT}, {T, X, Y, Z});
```

```
HilbertDimension({R, RX}, {T, X, Y, Z});
HilbertDimension({R, RY}, {T, X, Y, Z});
HilbertDimension({R, RZ}, {T, X, Y, Z});
HilbertDimension({RT, RX}, {T, X, Y, Z});
HilbertDimension({RT, RY}, {T, X, Y, Z});
HilbertDimension({RT, RZ}, {T, X, Y, Z});
HilbertDimension({RX, RY}, {T, X, Y, Z});
HilbertDimension({RX, RZ}, {T, X, Y, Z});
HilbertDimension({RY, RZ}, {T, X, Y, Z});
```

$$2$$
$$2$$
$$2$$
$$2$$
$$2$$
$$2$$
$$2$$
$$2$$
$$2$$
$$2 \tag{5}$$

> ***##Transversal intersection with {T=0}***

```
HilbertDimension({R, T, RX, RY, RZ}, {T, X, Y, Z});
Solve({R, T, RX, RY, RZ}, tdeg(Variables));
```

$$0$$

$$\{[[Y, Z, X, T], plex(T, X, Z, Y), :]\} \tag{6}$$

> ***##Transversal intersection with {Z=0}***

```
HilbertDimension({R, RT, RX, RY, Z}, {T, X, Y, Z});
Solve({R, RT, RX, RY, Z}, tdeg(Variables));
```

$$0$$

$$\{[[T, X, Y, Z], plex(Z, Y, X, T), :]\} \tag{7}$$

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